K.D.S.H.M. Perera

B.Sc. Engineering (Honours)

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Profile

As a Mechatronics Engineering graduate, I am driven by how biological designs can revolutionize robotics. My research focuses on the intersection of bio-inspired mechanisms and soft robotics, with a specific interest in the dynamic modeling and control of unconventional locomotion. I am particularly interested in how nature's morphological intelligence can be translated into robotic systems to enhance adaptability. Beyond individual platforms, I seek to scale these designs through swarm behavior to solve complex, decentralized tasks in unpredictable environments.

Education

2021 – 2025 **B.Sc. Engineering (Hons) in Mechanical Engineering**
Specialisation: Mechatronic Systems Engineering, *University of Moratuwa*, Sri Lanka
GPA: 3.79 / 4.00
Class Rank: 10/120
Minor: Mathematics

2017 – 2019 **GCE Advanced Level, Physical Science Stream**, *D.S. Senanayake College, Colombo 07*
Results: 2 A's & 1 B; Z-Score: 2.3409

2011 – 2016 **GCE Ordinary Level**, *D.S. Senanayake College, Colombo 07*
Results: 8 A's & 1 B

Work Experience

Feb 2025 – Present **Temporary Lecturer**, *Department of Mechanical Engineering, University of Moratuwa*, Moratuwa, Sri Lanka

Aug 2025 – Jan 2026 **Temporary Instructor**, *Department of Mechanical Engineering, University of Moratuwa*, Moratuwa, Sri Lanka

Mar 2024 – May 2024 **Intern / Student Researcher**, *Shandong University of Science and Technology*, Shandong, China

Nov 2023 – Feb 2024 **Intern / Project Engineer**, *Dipped Product PLC*, Sri Lanka

Academic Projects

Underwater Drone for Coral Observation Contributed to the design and development of a shallow-sea underwater drone using a novel fuzzy-adaptive sliding-mode controller for robust trajectory tracking; developed a VR-based teleoperation interface for real-time control and visualization.

Brachiating Robot Designed, simulated (Gazebo), and built a prototype brachiation robot to investigate dynamic locomotion and control across variable bar spacing and heights.

Environment Monitoring Station	Implemented real-time environmental monitoring with signal conditioning, processing, remote communication and data visualization (web/mobile).
App-Controlled Mobile Robot	Developed a differential-drive mobile robot with image-based object detection, path planning, and manual/autonomous operation via a mobile app.
Micro-ROS Based Home Automation	Led development of an IoT smart-home prototype integrating micro-ROS for autonomous temperature-based fan control and web-based lighting/fan management.
Smart Garbage Collector	Collaboratively engineered a mobile robot using image processing to autonomously navigate predefined routes and collect litter.
Modular Microconveyor	Designed a modular microconveyor using chevron electrothermal actuators and a serpentine beam ratchet for precise, incremental motion; validated with multiphysics simulation.

Industrial Projects

Dynamic Work Bar Transfer Mechanism	Designed a pneumatic system to halt and transfer work bars between leaching tanks.
Web-Based SCADA Visualization	Developed a full-stack web application for real-time SCADA dashboards and historical data visualization.
Hydraulic Lifting Platform	Researched and prototyped a hydraulic lift for wheelchair accessibility (Shandong University of Science and Technology).

Achievements

Best Paper Award	Robotics & Intelligent Systems Track — Moratuwa Engineering Research Conference (MERCon 2025), Moratuwa, Sri Lanka.
The Institution Best Project	Institution of Mechanical Engineers (IMechE) — University of Moratuwa, Sri Lanka (2025).
Dean's List	Semesters 1, 2, 5, 7

Research Publications

- (First author & presenter) *Brachiation Robot Inspired by Gibbon Dynamics: Insight for Control Across Irregular Bar Spacing and Heights*. Moratuwa Engineering Research Conference (MERCon 2025), Moratuwa, Sri Lanka. **Best Paper Award, Robotics & Intelligent Systems**.DOI: 10.1109/MERCon67903.2025.11217001.
- (First author & presenter) *A Fuzzy-Adaptive Sliding-Mode Control Strategy for Optimizing Control of an Underwater Drone*. MERCon 2025, Moratuwa, Sri Lanka.DOI: 10.1109/MERCon67903.2025.11217001.
- (First author) *Development of Marine Grade Acrylonitrile – Butadiene – Styrene (ABS) based Hull of an Underwater Drone*. MERCon 2025, Moratuwa, Sri Lanka. DOI: 10.1109/MERCon67903.2025.11217039.
- (First author) *CFD-Driven Hydrodynamic Optimisation of a Whale Shark–Inspired Underwater Drone for Shallow Reef Observation*. 12th International Conference on Heat Transfer and Fluid Flow (HTFF 2025). DOI:10.11159/htff25.243.
- (First author) *A Chevron-Actuated Microconveyor for Precise Wafer-Scale Microfactory Transportation*. Mechanical Engineering Research Symposium 2025 (MERS 2025), Department of Mechanical Engineering, University of Moratuwa. Accepted.

- (First author) *WS Ray: Extending Human Perception with a Teleoperated Underwater Drone for Subaquatic Environments*.International Conference on Automation Science and Engineering (CASE 2026). Under review.

Teaching & Mentorship Experience

Temporary Instructor/Lecturer	Assigned to design and deliver lectures on **Robot Operating System (ROS)** for Robotics modules and **Modeling & Simulation** for Advanced Mechatronics.
	Designed and delivered two workshops on ROS 2 for mechatronics undergraduates.
	Mentor two university teams developing an Autonomous Underwater Vehicle (AUV) for the Singapore AUV Challenge (SAUVC 2026).
	Conduct practical labs for Mechanical & Mechatronic Engineering modules.
Volunteer Experience	Volunteer as a mathematics tutor for local school children (in my native language, Sinhala) to support their education.

Skills

Programming	Python, C++, MATLAB & Simulink , LabVIEW
Robotics & Simulation	ROS2, micro-ROS, Gazebo, MuJoCo, Webots, OpenCV
Design & Analysis	SolidWorks, AutoCAD, Ansys, Adams, Fluidsim
Embedded Systems & Electronics	Atmel Studio, Proteus, Arduino & ESP32, Raspberry Pi
Web & Data	HTML/CSS, JavaScript, SQL, Flutter
Office & Graphics	LaTeX, Microsoft Word, PowerPoint, Excel, Adobe Illustrator, Photoshop

Languages

Sinhala	Native
English	Fluent

Co-curricular Activities

Committee Member	EXMO 2023 Logistics, University of Moratuwa
Committee Member	MechatronX 2023, Inter-School Robotics and Automation Competition
Society Memberships	IMechE, UMMES
School Activities	Science Quiz Team Organizer, D.S. Senanayake College
Community Service	Danuma Yathra Member (2019–2022)

References

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